ARCHIVES OF BIOCHEMISTRY AND BIOPHYSICS 140, 318–325 (1970)

Calcium Phosphate Formation *in Vitro*

II. Effects of Environment on Amorphous–Crystalline Transformation [1]

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Received February 9, 1970; accepted July 16, 1970

The amount of time that actually elapsed during the solution mediated transformation of amorphous calcium phosphate into crystalline apatite was short and invariable. On the other hand, the time required to reach this period of rapid precipitate conversion was strongly dependent upon the exact solution environment. Amorphous calcium phosphate was more stable when formed from solutions high in pH, ionic strength, and viscosity, rich in Ca^{2+} relative to HPO_4^{2-}, and low in initial supersaturation and temperature. The suspension lifetime of amorphous calcium phosphate was also sensitive to changes in dielectric constant. Amorphous calcium phosphate was also more stable in the presence of poly-L-glutamate, polyacrylate, phosvitin, casein, and large amounts of the protein–polysaccharides, or by the addition of small amounts of Mg^{2+}, $P_2O_7^{4-}$, CO_3^{2-} and F^-. Macromolecule-calcium phosphate flotation reactions (inhibition of ready sedimentation) were found for the casein, poly-L-glutamate, polyacrylate, and poly-L-lysine molecules as well as for the protein–polysaccharide complex.

Once initial mineral phase separation takes place during solution-mediated calcium phosphate formation, the next major event to occur is the transformation (via dissolution and reprecipitation) of the initial amorphous calcium phosphate precipitate into tiny crystals of apatitic calcium phosphate (1–7). The physicochemical mechanisms which govern this precipitate phase transformation process are such that once initiated it would normally proceed to completion, i.e., until only crystalline apatite is present (2–4). Therefore, amorphous calcium phosphate would be expected to have only a finite suspension lifetime, the natural extent of which may be dependent upon the exact experimental conditions used (2–4).

Bone tissue has always been found to contain substantial amounts of amorphous calcium phosphate as well as crystalline apatite (8–12). Thus, it is obvious that some factors must exist within living systems which regulate the physicochemical conversion of amorphous calcium phosphate to crystalline apatite. The variability and/or specificity of these factors is evidenced by the fact that while approximately 30% of the total mineral found within mature bone and dentine tissue is always amorphous calcium phosphate (8–12), no amorphous mineral has been found in mature dental enamel (8). Furthermore, only amorphous calcium phosphate has been found within isolated cell mitochondria which have been subjected to energy-dependent calcium loading in the

[1] This work was supported in part by U. S. Public Health Service Grant DE-01945 from the National Institute of Dental Research. Publication No. 67 from the Laboratory of Ultrastructural Biochemistry.

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presence of ATP and inorganic phosphate (13–15). Because of these findings and because both the physical and chemical properties of amorphous calcium phosphate have been found to be considerably different from those of crystalline apatite (2–5, 8, 11, 16), it is reasonable to suppose that within mammalian tissue systems, the amorphous–crystalline mineral phase transformation is subject to some form of biological control.

The results presented in this paper form part of an overall reexamination of calcium phosphate formation *in vitro* in which both crystallographic and kinetic considerations were carefully taken into account (7). This paper reports the manner in which the amorphous–crystalline precipitate phase transformation reaction is affected by a number of biologically pertinent environmental factors and outlines some kinetic and physical interactions between amorphous calcium phosphate and certain macromolecules.

MATERIALS AND METHODS

All solution, reaction, and sample-handling procedures used in this study were exactly the same as those already described in the first paper of this series (7). However, when initial Ca × P millimolar products greater than 50 mM² were used, aliquot centrifugation and precipitate recovery procedures described earlier (2) were substituted for those employed at lower levels of initial supersaturation (7). Amorphous–crystalline conversion was assessed by quantitative X-ray diffraction examination of the lyophilized precipitates (2, 9) and by following supernatant calcium concentration (2) using atomic absorption spectrophotometry (7). The chemical composition of the recovered precipitates was determined by methods described previously (7).

Macromolecular inhibition of the ready sedimentation of either amorphous or crystalline calcium phosphate precipitates (macromolecule-calcium phosphate flotation) was examined by differential centrifugation procedures (17, 18). The weight ratio of macromolecule to calcium phosphate in both the total precipitates and the different centrifugal fractions recoverable from these flotation reactions was determined from percentage moisture and percentage ash data (9). The buffer–organic solvent systems and their associated dielectric constant values utilized in this study were taken from previously published, tabulated data (19).

Reagent grade chemicals were used throughout. The commercial sources of the macromolecules used in this study were the same as those listed in the first paper of this series (7). In addition, histone-stripped soluble RNA (yeast) and preparations of L-α-lecithin and L-α-cephalin were obtained from General Biochemicals. The salt-soluble collagen, kindly provided by Dr. S. Rothbard of the Department of Medicine, Cornell University Medical College, New York, New York, was reconstituted and purified from young rat tail tendon. The protein–polysaccharide preparations used here were the same as those used earlier (7).

RESULTS

In every reaction examined in this study, the amount of time which elapsed from the very first detection of apatite crystals to the complete disappearance of the initial amorphous calcium phosphate precipitate did not vary from a value of 8 ± 3 min as long as pH was kept between values of 7.0 and 8.0 and the Ca × P millimolar product used for the initial reactant mixing was kept below a value of 150 mM². However, the amount of time which elapsed from initial solid phase separation to the start of this period of relatively rapid amorphous–crystalline conversion, i.e., the amorphous calcium phosphate suspension lifetime defined here as t_c, was strongly dependent upon the exact experimental conditions used. For example, Table I shows the effect of decreasing initial Ca^{2+} and HPO_4^{2-} concentration upon amorphous calcium phosphate suspension lifetime. As can be seen from this table, amorphous calcium phosphate appears to be much more stable if it is formed from solutions at low levels of initial supersaturation than if formed from solutions at relatively high initial supersaturation.

Throughout this study, solution conditions were generally chosen which afforded instantaneous precipitation at room temperature for the sake of experimental convenience. However, each effect reported in this paper was spot-checked by additional experimentation at 37° and at low levels of initial supersaturation in order to see if it was still valid at different operating conditions. Whenever the buffering capacity of

the Tris medium was found to be insufficient to maintain pH effectively throughout initial mineral-phase separation, a higher initial pH was always used at the time the reagents were mixed. Once amorphous calcium phosphate precipitation took place, the resultant pH value remained unchanged thereafter. Therefore, the pH values stated in this paper are those existing throughout the entire suspension lifetime and subsequent transformation of the initial amorphous phase.

Figure 1 shows the effect of initial Ca/P molar mixing ratio, while Fig. 2 shows the effect of pH upon amorphous calcium phosphate suspension lifetime (t_c). As can be seen from these figures, amorphous calcium phosphate stability is enhanced by solutions high in pH, rich in calcium, and poor in phosphate.

TABLE I

EFFECT OF INITIAL CONCENTRATION ON THE SUSPENSION LIFETIME OF AMORPHOUS CALCIUM PHOSPHATE[a]

Initial (Ca × P)	Suspension lifetime (min)
131.6 mM²	33 ± 2
50.6 mM²	42 ± 2
27.2 mM²	62 ± 2
21.0 mM²	104 ± 5
14.6 mM²	182 ± 6

[a] These reactions were conducted in Tris buffer, ionic strength = 0.15, pH = 7.40, with an initial Ca/P molar mixing ratio of 1.11 and constant gentle stirring at 25°. Initial Ca^{2+} and HPO_4^{2-} concentrations are expressed as initial Ca × P millimolar products.

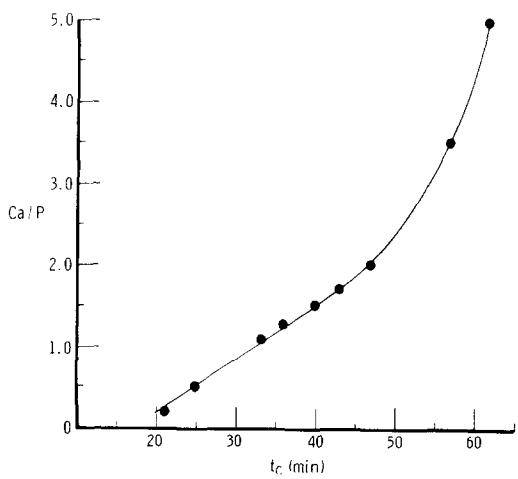

FIG. 1. Effect of initial Ca/P molar mixing ratio on amorphous calcium phosphate suspension lifetime (t_c). These reactions were all conducted in Tris buffer, ionic strength = 0.15, pH = 7.40, with an initial Ca × P product of 131.6 mM² and constant gentle stirring at 25°.

The effect of varying ionic strength (μ) and dielectric constant (ϵ) on t_c are shown in Figs. 3 and 4, respectively. These figures show that amorphous calcium phosphate is least stable when suspended in aqueous media of low dielectric constant and low ionic strength. In addition, we have found that amorphous calcium phosphate is 65% more stable in solutions whose calculated intrinsic viscosity has been increased to a value of 2.2 Cp by the addition of either 34% (w/v) glycerol or 26% (w/v) sucrose. The effect of temperature on amorphous calcium phosphate stability was even more striking. For example, at pH = 7.70, all other solution conditions being identical to those described for Figs. 3 and 4, amorphous calcium phosphate is three times less stable at 37°, while being 16 times more stable at

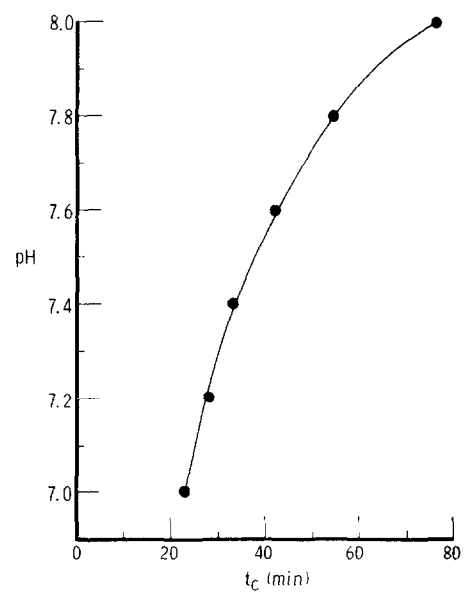

FIG. 2. Effect of pH upon amorphous calcium phosphate suspension lifetime (t_c). These reactions were all conducted in Tris buffer, ionic strength = 0.15, with an initial Ca × P product of 131.6 mM² and an initial Ca/P molar mixing ratio of 1.11 employing constant gentle stirring at 25°.

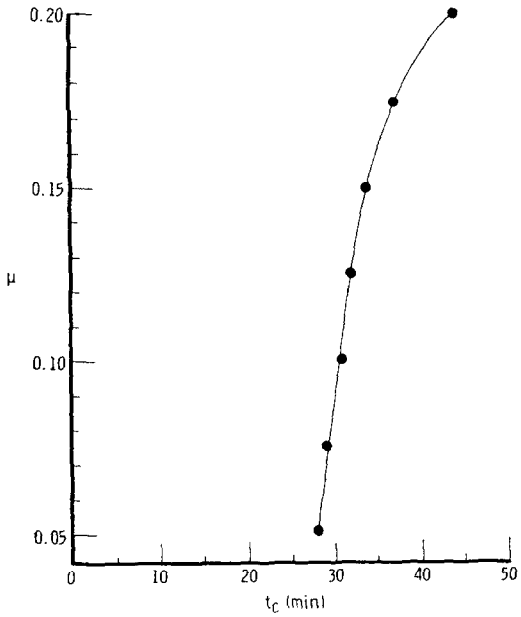

FIG. 3. Effect of varying ionic strength (μ) on amorphous calcium phosphate suspension lifetime (t_c). These reactions were conducted in Tris buffer, pH = 7.40, with an initial Ca × P product of 131.6 mM² and an initial Ca/P molar mixing ratio of 1.11 with constant gentle stirring at 25°.

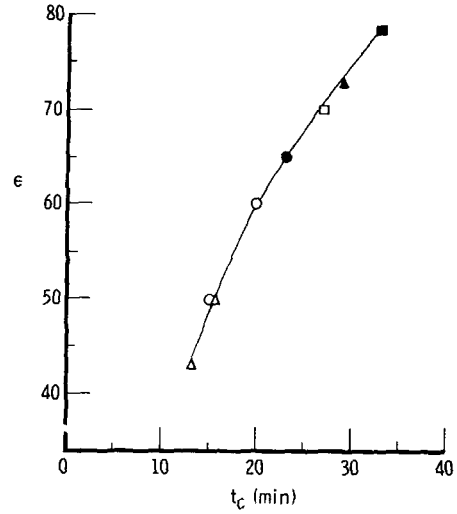

FIG. 4. Effect of varying dielectric constant (ϵ) on amorphous calcium phosphate suspension lifetime (t_c). These reactions were conducted in Tris buffer, ionic strength = 0.15, pH = 7.40, with an initial Ca × P product of 131.6 mM² and an initial Ca/P molar mixing ratio of 1.11 with constant gentle stirring at 25°. Dielectric constant values signified by open triangles were made by mixing buffer with *t*-butanol; those signified by open circles by mixing with dioxane; by closed circles with absolute ethanol and by closed triangle with ethylene glycol. The closed square is the control value.

4°, than at 25°. (This corresponds to t_c values of 16, 47, and 780 min for temperature values of 37, 25, and 4°, respectively.)

The effect of added ions on amorphous calcium phosphate suspension lifetime is shown in Table II. Sodium, potassium, and citrate were inactive in this test system even though these ions were added at three times their normal compact bone levels (20). As might possibly have been expected from previous investigations using higher concentrations of Mg^{2+} (21–23), CO_3^{2-} (24, 25), and $P_2O_7^{4-}$ (6, 26), we found all three of these ions effective amorphous calcium phosphate stabilizers even when used at very low relative concentrations. On the other hand, we were somewhat surprised to find that fluoride was an amorphous calcium phosphate stabilizer, since this ion has usually been thought of as an apatite-perfecting agent. The data of Table II show that the effect of these four stabilizing ions is additive when they are used in combination and that their relative strengths in stabilizing ability are as follows: $P_2O_7^{4-} > Mg^{2+} > F^- \gg CO_3^{2-}$.

The effect of added model macromolecules on amorphous calcium phosphate suspension lifetime is shown in Table III. Whenever a macromolecule was found to be insoluble under the conditions listed in the table, (e.g., polyglycine or poly-L-tyrosine) it was used in suspension. In all instances with the exception of the more concentrated protein–polysaccharide system listed in Table III, macromolecule-induced viscosity effects were negligible. Casein, phosvitin, Na polyacrylate, poly-L-glutamate, and the protein–polysaccharides all showed significant amorphous calcium phosphate stabilization activity. The relative stabilization strengths of these macromolecules were as follows: poly-L-glutamate ≫ Na polyacrylate > phosvitin, casein ≫ protein–polysaccharide. In light of the results obtained with the phosphoproteins, casein and phosvitin, the

possible involvement of bound organic phosphate in amorphous calcium phosphate stabilization was examined further by testing both solutions of sRNA and emulsions of phospholipids. In both cases, we observed considerable stabilizing activity. Amorphous calcium phosphate suspension lifetime was increased by more than 100% at initial concentrations of 1.0 and 0.2 mg/ml for the sRNA and the phospholipid preparations, respectively.

The protein–polysaccharide complex has been demonstrated to inhibit the ready sedimentation of freshly formed calcium phosphate precipitates (17, 18, 27) causing them to be split into two differently sedimentable centrifugal fractions. Therefore, we decided to determine whether a similar activity might also be demonstrated by other macromolecules and to examine any differently sedimentable centrifugal fractions that we might obtain for differences in mineral content, crystallographic phase, and amorphous–crystalline conversion rate. Of all the macromolecules used in this study (outside of the protein–polysaccharide complex), only casein, polyacrylate, poly-L-lysine, and poly-L-glutamate were found to inhibit the ready sedimentation of amorphous calcium phosphate and crystalline apatite. These data, summarized in Table IV, showed that in no case did any kinetic, percentage of ash, or crystallographic difference exist between either of the two differently sedimenting centrifugal fractions obtained from any of these macromolecular systems at any time during the course of the entire calcium

TABLE II

EFFECT OF ADDED IONS ON AMORPHOUS CALCIUM PHOSPHATE SUSPENSION LIFETIME[a]

Ion	Mole percentage added[b]	Suspension lifetime (min)
None	—	33 ± 2
Citrate	3.3	33
Sodium	15.0	33
Potassium	0.6	33
Fluoride	10.0	42
Fluoride	25.0	82
Magnesium	2.5	57
Magnesium	10.0	156
Carbonate	20.0	36
Carbonate	60.0	47
Pyrophosphate	1.0	47
Combination[c]	—	90

[a] These reactions took place in Tris buffer, ionic strength = 0.15, pH = 7.40, with an initial Ca × P product and Ca/P molar mixing ratio of 131.6 mM² and 1.11, respectively, with constant gentle stirring at 25°.

[b] Cations were added as Cl^- salts, anions as Na^+ salts. The mole percentage values stated are based upon the initial calcium concentration for cations and upon the initial phosphate concentration for anions.

[c] Na^+, K^+, citrate as above; F^- at 10 mole %; Mg^{2+} at 2.5 mole %; CO_3^{2-} at 60 mole %; and $P_2O_7^{4-}$ at 1 mole %.

TABLE III

EFFECT OF ADDED MACROMOLECULES ON AMORPHOUS CALCIUM PHOSPHATE SUSPENSION LIFETIME[a]

Macromolecule[b]	Macromolecule concn (mg/ml)	Suspension lifetime (min)
None	—	33 ± 2
Salt-soluble collagen	3.0	33
Gelatin	5.0	33
Protein–polysaccharide complex	5.0	36
Protein–polysaccharide complex	10.0	60
Chondroitin sulfate	3.0	38
Lysozyme	3.0	32
Casein	0.5	60
Phosvitin	0.5	70
Phosvitin	0.2	48
Polyethylene glycol	5.0	33
Polyvinylpyrrolidone	5.0	33
Na polyacrylate	0.5	86
Poly-L-lysine	5.0	32
Polyglycine	5.0	34
Poly-L-tyrosine	5.0	33
Poly-L-glutamic acid	0.5	1800

[a] These reactions were conducted in Tris buffer, ionic strength = 0.15, pH = 7.40, with an initial Ca × P product and Ca/P molar mixing ratio of 131.6 mM² and 1.11, respectively, using constant gentle stirring at 25°.

[b] The addition of equivalent concentrations of acrylic acid, lysine, or glutamic acid monomers had no effect on ACP suspension lifetime. We would like to thank Drs. J. C. Pita and D. S. Howell of the Department of Medicine, University of Miami School of Medicine, Miami, Florida, for suggesting the use of the higher protein–polysaccharide concentration shown above.

TABLE IV

MACROMOLECULE–CALCIUM PHOSPHATE FLOTATION DATA[a]

Macromolecule	R_1/R_2[b]	Weight ratio of macromolecule to calcium phosphate in:			
		Initial suspension	Total precipitate	Centrifugal fraction, R_1	Centrifugal fraction, R_2
Protein–polysaccharide complex	4.3	2.0	0.4	0.4	0.5
Casein	0.5	2.0	1.2	1.2	1.2
Na polyacrylate	0.1	5.0	0.4	0.4	0.4
Poly-L-lysine	10.7	5.0	0.3	0.3	0.3
Poly-L-glutamate	0.3	5.0	0.4	0.4	0.4

[a] In Tris buffer, ionic strength = 0.15, pH = 7.40, from reactions formed with an initial Ca × P product and an initial Ca/P molar mixing ratio of 131.6 mM² and 1.11, respectively, with constant gentle stirring at 25°. All macromolecules were added prior to initial precipitation.

[b] Weight ratio of precipitate recovered from centrifugation at 8500*g*-min (centrifugal fraction, R_1) to precipitate recovered from centrifugation of R_1 supernatant at 300,000*g*-min (centrifugal fraction, R_2).

phosphate formation reaction. Hence, this inhibition of ready sedimentation may best be described as a flotation effect. Further, the data of Table IV also indicate that this flotation phenomenon is associated with either the coexistence of both high molecular weight and high charge density within the same macromolecule (e.g., the protein–polysaccharide complex, poly-L-glutamate) or the ability of a given macromolecule to form aggregates or networks which might function in an exclusion process to differentially sediment calcium phosphate precipitates on the basis of overall particle size (e.g., casein).

DISCUSSION

The data presented in this paper demonstrate that the expected suspension lifetime of an amorphous calcium phosphate precipitate is strongly dependent upon the particular environment in which it resides. Inasmuch as initial mineral phase separation is also dependent upon solution environment (7), it can thus be seen that exact control of: (1) initial Ca^{2+}, HPO_4^{2-}, $P_2O_7^{4-}$, Mg^{2+}, F^-, CO_3^{2-} concentration; (2) pH, total ionic strength, viscosity, and temperature; and (3) the concentration of all macromolecular polyelectrolytes, will determine the complete sequence of events occurring within any given calcium phosphate-formation system.

The possible role of macromolecules in the tissue formation of calcium phosphate has been adequately reviewed elsewhere (20, 27–31). The findings of this and the previous investigation (7) would support the contention that macromolecules can influence both initial formation of amorphous calcium phosphate and the ultimate formation of apatite through amorphous–crystalline conversion.

The exact effect which a given macromolecule exhibits toward amorphous calcium phosphate formation is strongly dependent upon the initial degree of supersaturation of its calcifying solution (7). It may be possible that the reversal from activity as a phase separation inhibitor at relatively high initial supersaturation to that of a mineral phase separation catalyst at relatively low levels of initial supersaturation observed for casein (7), phosvitin (7), lysozyme (7), and the protein–polysaccharide complex (7, 27) may reflect macromolecular partitioning or sheltering of the amorphous calcium phosphate nucleation and/or growth process from total access to free solvent. For example, at high supersaturation, macromolecular effects would not be expected to compete efficiently with the relatively high driving forces for amorphous calcium phosphate formation existing within the solvent space itself. Thus, under these conditions, macromolecules probably behave only as Ca^{2+} or HPO_4^{2-} binding agents. However, at relatively low levels of initial supersaturation, i.e., where solution amorphous calcium phosphate formation lag times are quite long, specific macromolecules could contain sites either in their internal or solvation-shell space which may actually be favored locales for amorphous calcium phosphate nucleation and/or growth. Thus, under these latter conditions, amorphous calcium phosphate formation could take place within the protective

domain of the macromolecular species and not in free solution.

We did not find reconstituted collagen an effective inhibitor of amorphous–crystalline conversion, but as might have been expected from the results of others (28–35), we did find it an excellent initiator of amorphous calcium phosphate phase separation (7). The additional finding (7, 36) that gelatin can be an effective initial mineral phase separation catalyst would appear to indicate that native collagen conformation may not really be essential for this activity. However, the gelatin effect may not be so specific as the native collagen one since it occurs at higher relative concentrations.

The results obtained in this overall investigation with model macromolecules on initial mineral phase separation (7), amorphous–crystalline conversion, and calcium phosphate flotation provide ample evidence that macromolecules can definitely interact with calcium phosphate salts. Thus, these model system results are also compatible with previous suggestions that functional groups contained within acidic mucopolysaccharides such as the ester sulfate, glycuronic carboxyl, or *N*-acetyl groups (25–27, 37), charged amino acid side chains of proteins (28, 32, 36, 38–40), and seryl or threonyl ester phosphate on polypeptide chains (28, 41) might possibly serve as functional sites for matrix–mineral interactions within connective tissue.

Specifically, the data obtained here with phospholipids, RNA, and the phosphoproteins, phosvitin and casein, and with the free carboxyl-containing polymers, polyacrylate and poly-L-glutamate, indicate that organically bound phosphate and acidic amino acid side chains may very well be the most potent sites for any given macromolecule–mineral interaction. This view is reinforced by earlier apatite gel-chromatography data indicating that these two functional groups are the strongest receptor sites for the adsorption of proteins to calcium phosphate columns (42). However, the mild flotation result observed above for poly-L-lysine and the initial mineral phase separation catalysis activity found earlier (7) for both this polyamino acid and for lysozyme indicate that basic amino acid side chains could also serve as sites for macromolecule–mineral interaction. These latter findings are in agreement with previous data implicating basic amino acid residues as potential sites for collagen–mineral attachment within bone tissue (28, 39, 40) and suggesting that lysyl- and/or phosphoseryl-residues in casein may be the specific sites stabilizing amorphous calcium phosphate within casein–mineral micelles isolatable from bovine milk (43).

The results obtained in this investigation also indicate that more than just side chain characteristics may be involved in the interaction of a given macromolecule with the calcium phosphates. For example, we have found that macromolecular charge and size characteristics are important in both initial mineral phase separation catalysis (7) and macromolecule–mineral flotation reactions. We have also found that a decreased dielectric constant generally enhances both initial mineral phase separation (7) and amorphous–crystalline conversion. Thus, it is entirely possible that a local milieu favorable for either amorphous calcium phosphate or apatite formation might also be provided by a partially apolar region within a macromolecule as well as by its more polar regions. Consequently, macromolecular size and conformation should also be considered as possible contributing factors to any given macromolecule–mineral interaction.

REFERENCES

1. Watson, M. L., and Robinson, R. A., *Amer. J. Anat.* **93**, 25 (1953).
2. Eanes, E. D., Gillessen, I. H., and Posner, A. S., *Nature London* **208**, 365 (1965).
3. Eanes, E. D., and Posner, A. S., *Trans. N. Y. Acad. Sci.* **28**, 233 (1965).
4. Eanes, E. D., Gillessen, I. H., and Posner, A. S., *in* "Crystal Growth" (H. S. Peiser, ed.), p. 373. Pergamon Press, New York (1967).
5. Walton, A. G., Bodin, W. J., Furedi, H., and Schwartz, A., *Can. J. Chem.* **45**, 2695 (1967).
6. Francis, M. D., *Calcif. Tissue Res.* **3**, 151 (1969).
7. Termine, J. D., and Posner, A. S., *Arch. Biochem. Biophys.* **140**, 307 (1970).

8. TERMINE, J. D., AND POSNER, A. S., *Science* **153,** 1523 (1966).
9. TERMINE, J. D., AND POSNER, A. S., *Calcif. Tissue Res.* **1,** 8 (1967).
10. TERMINE, J. D., WUTHIER, R. E., AND POSNER, A. S., *Proc. Soc. Exp. Biol. Med.* **125,** 4 (1967).
11. TERMINE, J. D., PULLMAN, I., AND POSNER, A. S., *Arch. Biochem. Biophys.* **122,** 318 (1967).
12. QUINAUX, N., AND RICHELLE, L. J., *Isr. J. Med. Sci.* **3,** 677 (1967).
13. GREENAWALT, J. W., ROSSI, C. S., AND LEHNINGER, A. L., *J. Cell. Biol.* **23,** 21 (1964).
14. WEINBACH, E. C., AND VON BRAND, T., *Biochim. Biophys. Acta* **148,** 256 (1967).
15. THOMAS, R. S., AND GREENAWALT, J. W., *J. Cell. Biol.* **39,** 55 (1968).
16. EANES, E. D., TERMINE, J. D., AND POSNER, A. S., *Clin. Orthop.* **53,** 223 (1967).
17. SACHS, C., AND SCHUBERT, M., *Science* **142,** 1073 (1963).
18. DI SALVO, J., AND SCHUBERT, M., *J. Biol. Chem.* **242,** 705 (1967).
19. HARNED, H. S., AND OWEN, B. B., *in* "Physical Chemistry of Electrolytic Solutions," Ed. 3, p. 161. Reinhold, New York (1958).
20. MCLEAN, F. C., AND URIST, M. R., "Bone," Ed. 3, pp. 56, 72. Univ. of Chicago Press, Chicago (1968).
21. BOULET, M., MARIER, J. R., AND ROSE, D., *Arch. Biochem. Biophys.* **96,** 629 (1962).
22. BACHRA, B. N., TRAUTZ, O. R., AND SIMON, S. L., *Arch. Oral Biol.* **10,** 731 (1965).
23. EANES, E. D., AND POSNER, A. S., *Calcif. Tissue Res.* **2,** 38 (1968).
24. BACHRA, B. N., TRAUTZ, O. R., AND SIMON, S. L., *Arch. Biochem. Biophys.* **103,** 124 (1963).
25. NEUMAN, W. F., AND MULRYAN, B. J., *Calcif. Tissue Res.* **1,** 94 (1967).
26. FLEISCH, H., RUSSELL, R. G. G., BISAZ, S., TERMINE, J. D., AND POSNER, A. S., *Calcif. Tissue Res.* **2,** 49 (1968).
27. SCHUBERT, M., AND PRAS, M., *Clin. Orthop.* **60,** 235 (1968).
28. GLIMCHER, M. J., AND KRANE, S. M., *in* "Treatise on Collagen" (B. S. Gould, ed.), Vol. 2B, p. 67. Academic Press, New York (1968).
29. BACHRA, B. N., *Clin. Orthop.* **51,** 199 (1967).
30. FLEISCH, H., *Clin. Orthop.* **32,** 170 (1964).
31. SOBEL, A. E., BURGER, M., AND NOBEL, S., *Clin. Orthop.* **17,** 103 (1960).
32. URIST, M. R., AND ADAMS, J. M., JR., *Arch. Pathol.* **81,** 325 (1966).
33. RAMACHANDRAN, G. N., AND SANTHANAM, M. S., *Experientia* **12,** 340 (1956).
34. WADKINS, C. L., *Calcif. Tissue Res.* **2,** 214 (1968).
35. BACHRA, B. N., AND FISCHER, H. R. A., *Calcif. Tissue Res.* **2,** 343 (1968).
36. WALTON, A. G., FRIEDMAN, B. A., AND SCHWARTZ, A., *J. Biomed. Mater. Res.* **1,** 337 (1967).
37. BOWNESS, J. M., AND LEE, K. H., *Biochem. J.* **103,** 382 (1967).
38. SCHIFFMAN, E., LAVENDER, D. R., MILLER, E. J., AND CORCORAN, B. A., *Calcif. Tissue Res.* **3,** 125 (1969).
39. SOLOMONS, C. C., AND IRVING, J. T., *Biochem. J.* **68,** 499 (1958).
40. WUTHIER, R. E., COTMORE, J. M., AND MARON, S. S., *Calcif. Tissue Res.* **1,** 288 (1968).
41. VEIS, A., AND PERRY, A., *Biochemistry* **6,** 2409 (1967).
42. BERNARDI, G., AND KAWASAKI, T., *Biochem. Biophys. Acta* **160,** 301 (1968).
43. ROSE, D., AND COLVIN, J. R., *J. Dairy Sci.* **49,** 351 (1966).